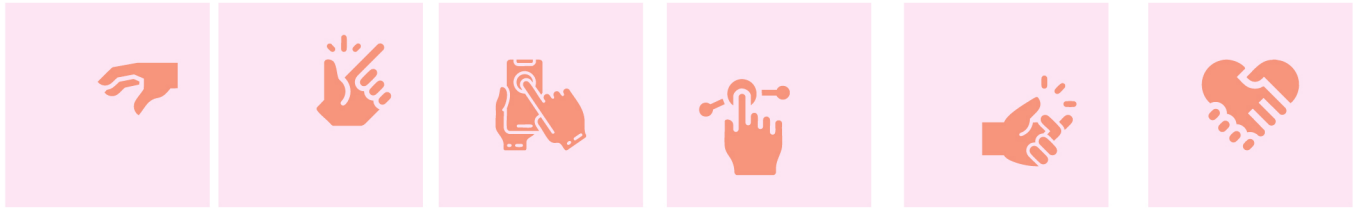
UX Portfolio

User Research | User Experience



Nishabh Polke

01

Usability Testing for a Smart Home IoT Application

2022 | UX Research | User Testing

Project Overview

This project aimed to **evaluate the usability of a mobile application designed to unify IoT product users on a single platform**, offering a seamless smart and connected home experience. The application allows users to manage smart devices such as lights, fans, and more, enabling them to assign devices to specific homes and rooms for efficient control. The client, who developed the application, sought to understand its usability across diverse user groups and identify errors, needs, and wants associated with such an application. The study focused on metropolitan cities in India, where the project team conducted user testing to gather insights from real-world users.

My Role

Led a user testing study, including the creation of discussion guides and identification of key testing objectives to evaluate the application's usability.

User Research Objectives

1. Can users navigate the system intuitively without requiring external assistance?
2. Do the connotations effectively convey the feature and its intended outcome in a way that is clear and easily understandable?
3. Can users navigate the system intuitively without requiring external assistance?
4. How easy is it to use across various use cases?
5. What additional use cases or features would users find valuable?
6. Which features of the app are users most and least likely to use?

User Research Goals

1. The research must focus on engaging with the right audience across the country to gather valuable insights through observations and interviews.
2. The research should be designed to uncover the implicit needs and demands of this consumer, who may not readily express them.

Research Methodology

Research Objectives, Techniques and Sample Planning

Respondent Recruitment + Discussion Guide preparation

Pilot Interviews + Discussion Guide Refinements.

Fieldwork/ Data Collection; Data Cleaning

Analysis and synthesis of research Findings

Presentation of research Findings and Interfaces.

Research Techniques

Initiation

- Install App in their device
- Start screen recording
- Let the users use the application as per their convenience.
- Help only if asked for

Observe

- Navigation Paths
- Unsaid Painpoints
- Errors made
- Success /failure rates

Post-use Interview

- Assess users' views on the app's ease of use and its value.
- Evaluate the clarity of symbols, icons, and overall communication.
- Identify features that positively engage or surprise users.

Research Study

Objective

Identify user pain points in app navigation.

Observe and gather insights on app usability and value.

Understand user sentiments and gather feedback on improving the app experience

Analyze user data to generate UX improvement suggestions and present them to clients to guide thoughtful design iterations.

Fieldwork

Where?

Delhi
Mumbai
Pune
Bangalore
Coimabtore
Kochi

Who?

- 10 usability tests were conducted in each city per week (5 Female & 5 Male) .
- All the interviewees belonged to the SEC AI category which would resonate with the type of potential users for the application.

How?

Throughout the weeks, the research team visited each respondent's home.

The research agenda was briefly explained to the interviewees, and their consent to record data was requested.

Participants received a device with the App pre-installed and were asked to complete a series of tasks.

If participants needed help, the team offered direction and cues to navigate the tasks.

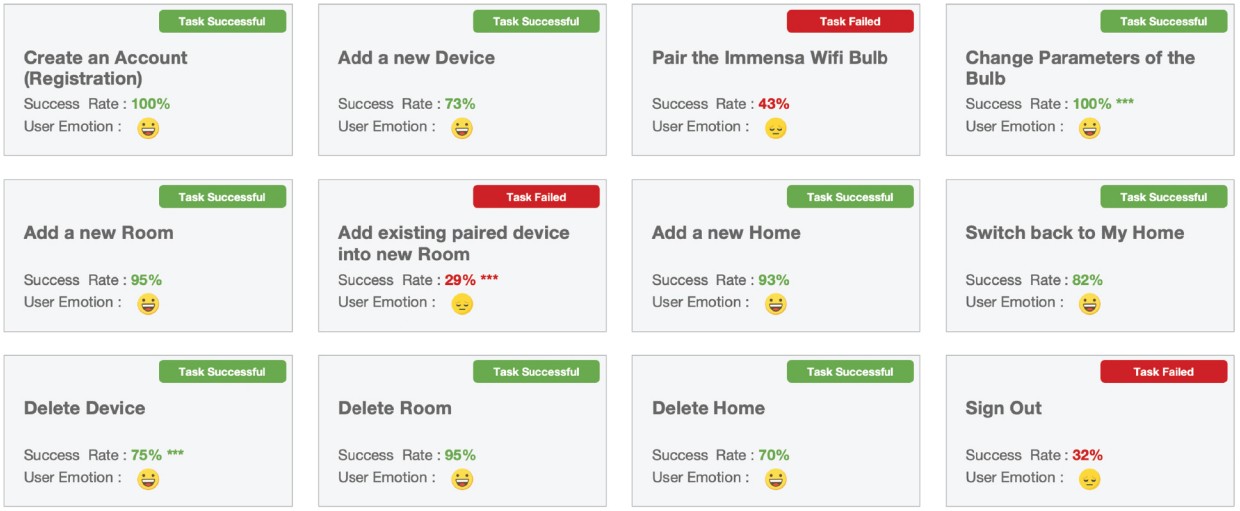
Following the user testing session, participants were interviewed to capture their experience and any feedback they had.

Task Analysis

Twelve key tasks were identified within the app that were critical to evaluating its overall success.

Data recording process:

- An Excel sheet was maintained with all tasks listed. A binary system was used to track task performance — a successful completion was marked as '1', while a failed attempt was marked as '0'.
- Alongside this, user pain points were documented in the same sheet. These were captured through observation and by noting users' verbal feedback as they performed each task.



Data recording process:

An Excel sheet was maintained with all tasks listed. A binary system was used to track task performance — a successful completion was marked as '1', while a failed attempt was marked as '0'. Alongside this, user pain points were documented in the same sheet. These were captured through observation and by noting users' verbal feedback as they performed each task.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1		**Register**	**Add a Device** - Add(+) a device - Select Category - Select Device - Connect with WiFi - Account Verification	**Pairing the smart bulb** - Add device name	**Change Parameters of the Bulb** - Colour - Mode - Brightness	**Add new home**	**Add a room**	**Add existing paired device in the room**	**Switch back to MyHome**	**Change room setting** - Room name - Room Location - Wallpaper	**Change Device settings** - Device Name - Device Location	**Delete Device**	**Delete Room**	**Delete house**	**Login**
2	.														
3	**User 1: Jacky Trivedi, 35**														
4	Time taken to complete task.....														
5	Successful completion of task (1=Success , 0=Fail)	1	1	1*	1	0	1	0	1	1	1		1	1	0
6	Observations related to difficulties	the user used the mobile phone number option	the user had to enter his email ID. Had difficulty finding the OTP.	the user pressed confirm without checking the instructions. He tapped on the infographics. He had to be instructed to use the device step by step. The instructions weren't clear for the user to proceed.											
7	Remarks				there was a lag on the app which left the user confused.	the user went to the hamburger menu > other to add the room. He had to be guided and nudged to find the option.	saving the changes led to the device pairing screen. Back button didn't work while testing.								This task was not carried out with this user.
8															
9	**User 2: Manali Garg,35**														
10	Time taken to complete task														
11	Successful completion of task (1=Success , 0=Fail)	1	1	0	0	1	1	0	1	1	1	0	1	1	0
12	Observations related to difficulties	the user took time to understand the instructions of password settings													
13	Remarks	the user used the phone number option to login. The application crashed while doing the task. User had to start from scratch. The user was not able to go through this process smoothly.	the Email id had to be registered again.	user pressed confirm directly. She was asked to repeat the task. The user didn't go through the on and off process of pairing the bulb.		the user had to be guided through the procedure.Finding the location became hard for the user. She used a dummy one for the sake of the interview.		the device was not connected and hence the task was not carried out.				The user was not able to connect the device.		the user first went to the drop down menu in the add home section to delete the home.	The user did not complete the registration process and was trying to login. She failed to login and had to start the registration process again.
14															
15	**User 3: Kajal Jadhav, 36**														
16	Time taken to complete task														
17	Successful completion of task (1=Success , 0=Fail)	1	0	0	0	1	1	0	1	0	0	0	1	1	0
18	Observations related to difficulties	the user used the email id option to login to create the account.		The user didn't follow through the instructions of pairing the device.											
19	Remarks	The user had difficulty creating a password with all the instructions. The user couldn't find the password sent on the mail. They had to be guided to check the spam folder.			The user failed to set up the device. This task could not be conducted.	the user used the add button to add a new home. She then later figured out how to add a home.		The user failed to set up the device. This task could not be conducted.		the user struggled to find the my home settings to change the parameters.		The user failed to set up the device. This task could not be conducted.			The user was not able to sign out of the account.
20															
21	**User 4: Purvi Shah,37**														
22	Time taken to complete task														
23	Successful completion of task (1=Success , 0=Fail)	1	1	1*	1	1	1	0	1	1	1	1	1	1	0
24	Observations related to difficulties	The app crashed while registering. The user then switched to entering an email option to register as the app kept crashing.		the user skipped through reading the information present on the screen. They did not add the device successfully.		The user struggled to find the option.		the user didn't understand where to find the paired device to change its location.						the user went to the edit room to delete the house. She tried deleting the rooms for this task. She was then able to navigate to manage homes to delete the task.	She wasn't able to find the logout option. She went to the help section. She gave up on finding how to logout of her account.
25	Remarks	The user used the phone number option to register. The user was bale to find the OTP in the spam folder.		The user had to be guided towards pairing the device. They did not read the instructions at first.		the user didn't understand the purpose of the wallpaper setting.		The user had to be guided to accomplishing the task.							
26															
27	**User 5: Arpit, 30**														
28	Time taken to complete task														
29	Successful completion of task (1=Success , 0=Fail)	1	1	1	1	1	1	0	1	1	1	1	1	1	1
30	Observations related to difficulties			the user switched it on and off only 3 times and waited for the device to start blinking. He then figured it out on his own to pair the device		the user couldn't pin point their location so they used a random location for the sake of the interview.		the user was not able to shift the device to the other room.							the user was able to figure out the login process.
		the user used the phone number option to	the user went into the hamburger option	the user read through the	the user experienced a lag while	the user went into the hamburger option		the user started adding the			the application crashed				the user tapped on the notifications

Mumbai | Ahemdabad | Bangalore | Kochi | Coimbatore | Delhi | +

Researcher's Notes : Data recorded from each user test on the spot

Research Study - Findings & Presentation

Each task was analyzed using a binary system, and the resulting statistics were presented to the client. For tasks where the majority of users were unsuccessful, the underlying reasons were identified and discussed. Additionally, recommendations were provided—grounded in established Heuristic Principles—to enhance the application's overall user experience.

Task 3 : Pair the Immensa Bulb

Task Failed

Instructions :The users were asked to read the instructions presented to them on the screen and pair the device accordingly

Completion rate : 43 %

Easy — (Perception Indicator) — Difficult

Number of users who completed task successfully

20 / 60

6 / 60 ***seeked assistance / Were helped

Number of users who failed to perform the task

34 / 60

Difficulties faced by the users:

- Most users did not clearly understand the instructions provided and proceeded to press "Confirm" and "Continue" before trying to pair the device.
- Users also clicked on the Infographics instead of trying to switch the bulb on & off from the main switch since the infographics seemed clickable to them. They asked for help at this point and had to be guided.
- Some users also mentioned that the font size of the instructions was too small and therefore didn't bother reading it.
- A majority of the users needed some guidance on how to pair the device. They were confused while looking at the instructions and what it meant, and asked us to explain it to them in detail

Link to Task Responses sheet:
https://docs.google.com/spreadsheets/d/15Zf42Sxk7J3pqMtkEbUS9EVhF39d6iu1bt-QsEczbh8/edit?usp=sharing

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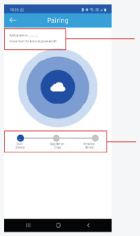
Task 3 : Pair the Immensa Bulb



Visibility of system status

The sense of control can be evoked by providing information about the system status and feedback after every interaction.

Suggestion : The details of the bulb (name, room name in which the device is being added) should be shown here to assure the user that the action being performed is correct.

Help Users Recognize, Diagnose, And Recover From Errors

Here the user is not informed that the pairing is unsuccessful (if that happens).

Suggestion : An error message can be displayed here if the pairing is unsuccessful so that the user knows that the process needs to be carried out again.

Visibility of system status

The user clicks on the power button but the confirmation that the bulb is turned on/ off on the screen is missing.

Suggestion : If the bulb were to light up here on the screen, the user would get a feedback and confirmation that the device has been turned on.

Link to Task Responses sheet:
https://docs.google.com/spreadsheets/d/15Zf42Sxk7J3pqMtkEbUS9EVhF39d6iu1bt-QsEczbh8/edit?usp=sharing

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Task 3 : Pair the Immensa Bulb

Appears like a wall of text.

These infographics seemed clickable.

The users clicked on these buttons to move on to the next step rather than pairing the device and clicking on them as it was intended.

Flexibility And Efficiency of Use

Declutter the screen and make the app easier to navigate. The app should always display only relevant UI elements and commands.

Suggestion : The information can be displayed in parts on swipeable screens which would display each step with an GIF. This would help reduce the cognitive load for the user.
The confirm and continue buttons could be incorporated on the last screen.

Link to Task Responses sheet:
https://docs.google.com/spreadsheets/d/15Zf42Sxk7J3pqMtkEbUS9EVhF39d6iu1bt-QsEczbh8/edit?usp=sharing

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Task 7 : Add a new home

Task Successful

Instructions :The users were presented a scenario and were asked to find a way to create a new home in the app

Completion rate : 93 %

Easy — (Perception Indicator) — Difficult

Number of users who completed task successfully

56 / 60

Number of users who failed to perform the task

04 / 60

Difficulties faced by the users:

- A lot of the users thought that the "others" option in the "Add Room" section meant it would add a home. They had to explore more in order to find the correct "Add home" button.
- While trying to add a new home, users had difficulty selecting a location since there was no other option but to pinpoint the location on the screen which they found hard.
- In case of mandatory fields that needed to be filled, validations weren't provided therefore the users didn't find the need to fill them and had to be prompted to do so.

Link to Task Responses sheet:
https://docs.google.com/spreadsheets/d/15Zf42Sxk7J3pqMtkEbUS9EVhF39d6iu1bt-QsEczbh8/edit?usp=sharing

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Task 7 : Add a new home

Consistency And Standards

A comprehensible system should never confuse users by using different words, visuals, or actions for the same concepts.

Suggestion : The stardalone word 'other' is interpreted differently by different users. 'Add another room' or 'new room' would give more clarity to the user.

Recognition Rather Than Recall

A good user interface doesn't require the user to recall frequently. Instead, it offers all options and information required to make a choice.

Suggestion : Here the user is forced to recall the location and drag the pin around to the exact location which can be frustrating. Instead the user should be given the freedom to type in the location and the pin gets redirected to typed in location.

Link to Task Responses sheet:
https://docs.google.com/spreadsheets/d/15Zf42Sxk7J3pqMtkEbUS9EVhF39d6iu1bt-QsEczbh8/edit?usp=sharing

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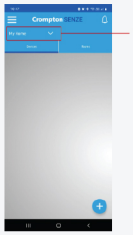
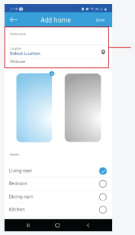
Task 7 : Add a new home



The Match Between The System And The Real World

People are approaching every new system with a mental model in mind. In other words, people presume how the system could work based on their experience with other systems that are similar.

Suggestion : Here a home icon could be included to help the user recognize this feature through graphic representation.



Error Prevention

The user misunderstands the communication and consciously performs an action which leads to a different result than they intended.

Suggestion : Here the user can be nudged to type in the text and enter the location with the text – *"required."* This will help the users understand that this is a mandatory field and it must be filled in order to save a home.

Link to Task Responses sheet:
https://docs.google.com/spreadsheets/d/15Zf42Sxk7J3pqMtkEbUS9EVhF39d6iu1bt-QsEczbh8/edit?usp=sharing

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A detailed interview was conducted after users completed all assigned tasks. Participants were asked questions regarding usability, current preferences, and future expectations. Each interview was recorded for research purposes and later transcribed into a separate Excel sheet. The transcriptions were analyzed to identify recurring patterns and insights within the sample group.

When you install a new application, what is your most preferred login option ? Via Email Address or Phone number ? Why ?

Preference :

43 / 60
Prefer registering with Phone Number

Reasons :

- Quick and easy to access
- Security
- No spam emails

10 / 60
Prefer registering with Email ID

Reasons :

- Personal preference
- Feeling of security with this option
- Flexibility in terms of unsubscribing than receiving spam calls

07 / 60
Do not have a Preference

Link to Questionnaire Responses sheet:
https://docs.google.com/spreadsheets/d/1mo7uwFl3l5x8dztmZ5-O8AlOCXurykGl7KzUkVlVX8/edit#gid=1854961073

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Do you think the information provided along with the image/icon was understandable? If not, what other information do you think should be accompanied with the device image?

Most Common Answer:

- The information provided on the screen was sufficient and easy to understand.

75% Of the users agreed that the information provided on the screen was sufficient for them to navigate through the application.

25% Of the users though that there should have been more information which would help them use the application.

Observations :

- The pairing screen had the most number of instructions present.
- Most of the users read through the instructions > pressed confirm > continue. They did not perform the task of switching the bulb on-off 3 times. They thought that the 'confirm button' was to validate if they have read the instructions or not.
- The instructions present on the screen are not accurate. Eg. The user has to perform the action of switching the bulb on-off a 4th time for the bulb to go into pairing mode. Almost all the users who performed this step had to be told to do the action one more time than what is suggested.
- Some users clicked on the infographics present on the screen to pair the device.

Link to Questionnaire Responses sheet:
https://docs.google.com/spreadsheets/d/1mo7uwFl3l5x8dztmZ5-O8AlOCXurykGl7KzUkVlVX8/edit#gid=1854961073

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After using this application, do you think something could've been done differently that could ease the process or the task?

User Opinions :

"For a first time user its difficult..उसर कोई बात समझ में नहीं आती है तो there should be some kind of help..Chatting का provision हो उसर from the company"

Chatbot and 24x7 Customer care should exist to assist the users at any point of time.

"When I looked at it the first time, it's not looking so attractive. Just blank, so much of blank space. So I get confused as to what will happen.Something lively/anything moving like animations should exist "

"Visuals are not pretty enough"

"Background can be a little catchy. The light background is not so attractive."

Making the application more aesthetically pleasing.

"The pairing, I thought I had to do it from the phone.So if you could just add that I have to do it on the switch"

More accurate information should be displayed for carrying out a particular task.

"Instead of the wallpaper, I could have added a real image which would be like a personal touch"

The option to customize the wallpaper should exist which would generate more familiarity with the spaces of the house in the app.

Link to Questionnaire Responses sheet:
https://docs.google.com/spreadsheets/d/1mo7uwFl3l5x8dztmZ5-O8AlOCXurykGl7KzUkVlVX8/edit#gid=1854961073

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02

Usability Testing for a Tobacco Cessation Application

2022 | UX Research | User Testing

Project Overview

The Tobacco Cessation Application is designed for users aged 18 and above who are motivated to quit tobacco use—whether smoked or smokeless. The app's key features include a chatbot and a personalized quit plan, both aimed at supporting users in their cessation journey. Additionally, users have the option to connect with tobacco cessation experts for one-on-one guidance sessions.

User testing was conducted across four Indian cities, where the team traveled to engage with participants directly. In-depth interviews were carried out, recorded, and supplemented with detailed observational notes to capture insights.

The collected data was then analyzed and presented to the client, **offering analytical insights into the application's usability and overall effectiveness.**

My Role

Conducted a comprehensive user testing study, which involved creating discussion guides, conducting user interviews, and generating detailed reports based on the collected data.

User Research Objectives

1. To identify the features and content most appreciated by users in terms of usefulness and engagement.
2. To assess the level of awareness about nicotine among the general public.
3. To evaluate the clarity and effectiveness of the language in conveying the features and their intended outcomes.
4. To assess users' ability to navigate the application independently.
5. To assess the ease of use across different use cases.
6. To identify additional use cases (features) that users would like to have.
7. To determine the most and least frequently used features of the app by users.
8. To evaluate the usefulness and utilization of the appointment feature..
9. To identify the current pain points experienced by users in the app flows.

User Research Goal

1. To evaluate the usability and perceived usefulness of the application among diverse user groups with varying tobacco use preferences.
2. To gather insights across age, gender, and type of tobacco use, identifying patterns in user preferences and ease of use.
3. To generate insights that inform targeted design enhancements aligned with user needs and app goals.

Research Methodology

Research Objectives, Techniques and Sample Planning

Respondent Recruitment + Discussion Guide preparation

Pilot Interviews + Discussion Guide Refinements.

Fieldwork/ Data Collection; Data Cleaning

Analysis and synthesis of research Findings

Presentation of research Findings and Interfaces.

Research Techniques

Initiation

- Provide users with a device preloaded with the application.
- Initiate screen recording before the session begins.
- Allow users to explore and use the application at their own pace.
- Assign specific tasks for users to complete during the session.
- Offer assistance only upon user request, to preserve the authenticity of the experience.

Observe

- **Navigation Patterns:** Common paths taken by users to complete tasks.
- **Unspoken Pain Points:** Implicit challenges or frustrations not verbally expressed.
- **Hesitation Zones:** Sections where users are reluctant to share information.
- **User Errors:** Mistakes made while interacting with the application.
- **Task Outcomes:** Success and failure rates in completing assigned tasks.

Post-use Interview

- Understand users' perceptions of the usability and usefulness of the app's features in addressing quitting challenges.
- Gain insights into the struggles users face while quitting and assess how the app helps overcome these barriers and provide support.
- Evaluate the ease of understanding, focusing on the clarity of communication, symbols, and icons.
- Explore moments of delight that enhance the user experience.

Research Study

Objective

Identify user pain points in app navigation.

Observe and gather insights on app usability and value.

Understand user sentiments and gather feedback on improving the app experience

Analyze user data to generate UX improvement suggestions and present them to clients to guide thoughtful design iterations.

Fieldwork

Where?

Pune
Delhi
Lucknow
Nashik

Who?

- A total of 100 interviews were conducted. 25 users in each city.
- The sample consisted of 54 males and 46 females. Among them, there were 22 SLT users and 13 smokers from each gender, along with 19 male and 11 female participants who used both SLT and smoked.

Gender	Tobacco Usage	A	B	C	D	E	F	TOTAL
F	SLT	1	4	7	7	3		22
	Smokers	2	1	4	4	2		13
	Smoking + SLT	2	6	1	2			11
F total		5	11	12	13	5		46
M	SLT	6	3	4	4	1	4	22
	Smokers	1		4	6	2		13
	Smoking + SLT	8	4	2	1	2	2	19
M total		15	7	10	11	5	6	54
	TOTAL	20	18	22	24	10	6	100

F - Female
M- Male

Age Category
A : 20-25
B : 26-30
C : 31-35
D : 36-40
B : 41-45
C : 46-50

SLT - Smokeless Tobacco

How?

Participants were invited to a designated venue in each city for the user testing session.

Participants were asked to freely explore the app, navigate through its features, and read the informational content provided.

Interview 1: Participants were interviewed with a focus on the usability of the app, particularly regarding its convenience and ease of navigation.

After the first interview, participants were guided to use the "Book Appointment" feature within the app and schedule a short session for themselves.

Participants engaged in a 15-minute consultation with an online expert through the app.

Post-consultation, participants filled out a Google Form to provide quantitative data related to their preferences and experiences.

Interview 2 : Participants were interviewed again to gather feedback on the usefulness of the content and information provided within the app.

Interview 2 : A final set of questions was asked to assess the desirability of the app and to gather suggestions for making it more engaging and effective.

Research Study : Summary

What features and content do users find most useful and engaging, especially for nicotine awareness?

While most users initially said they **liked the 'Create a Quit Plan'** feature, interviews revealed a stronger preference for the **'Resources' section.**

Awareness around NRT is low, leading to several misconceptions among users.

Are the terms clear and do they convey the feature and its outcome effectively?"

Users are able to easily understand the meaning and purpose of the icons, as well as the accompanying text, indicating that **the visual elements and labels are intuitive and effectively designed.**

Can users navigate the platform easily without external assistance?

The majority of users navigated the application without any difficulty.

A few faced challenges due to limited experience with smart devices.

75% of users were able to navigate the app easily.

How easy is it for users to complete different tasks or use cases?

Ease	Task
86 %	Sign-in
75 %	Create a Quit Plan
73 %	Book an Appointment
79 %	Interact with Chatbot
73 %	Consumption Log
75 %	Navigation

On average, users found the application **77% easy to use.**

Research Study : Summary

What additional features or use cases would users like to see?

- Enhancing language clarity and accessibility
- Improving the user experience of the chatbot
- Upgrading the stories section
- Incorporating success stories
- Adding notifications as reminders
- Offering more flexibility for booking or rescheduling appointments"

Which features of the app do users like the most and the least?

Most Liked Feature - **58%**
'Create a Quit Plan'

Least liked feature - **22%**
'Exploring Content'

However, insights from user interviews indicate the contrary.

The usefulness and frequency of use of the appointment feature.

97% of users felt satisfied after speaking with an expert, and **89% expressed interest in doing so again.**

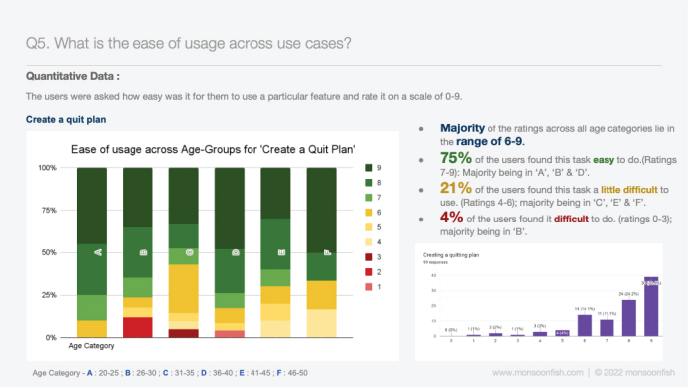
Though initially hesitant, many users described the experience as comforting—like talking to a trusted friend.

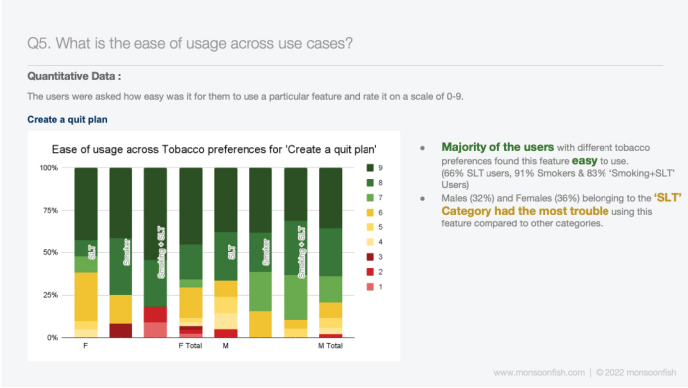
What are the current pain points users experience with the app's flow?

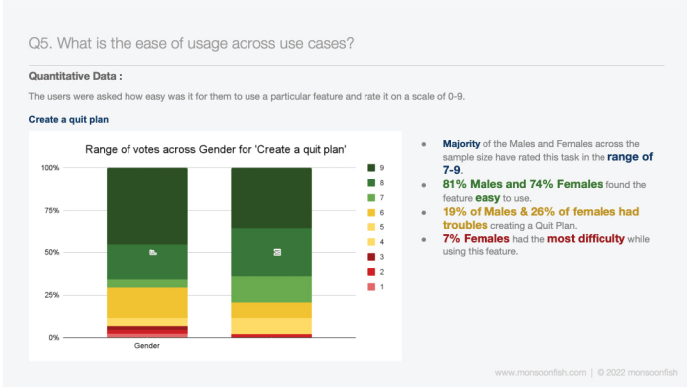
- Overly long text messages
- Insufficient time to read through stories
- Technical issues occurring after network loss
- Limited options for filtering content
- Difficulty uploading pictures
- Lack of system feedback to help users identify and resolve errors

Qantitative Data : Task Analysis - Presentation

Each task was analyzed based on user age, gender, and tobacco usage preferences. For example, in the 'Create a Quit Plan' task, pivot tables were used to assess ease of use across all categories, helping to identify specific problem areas. This approach was consistently applied to all tasks.



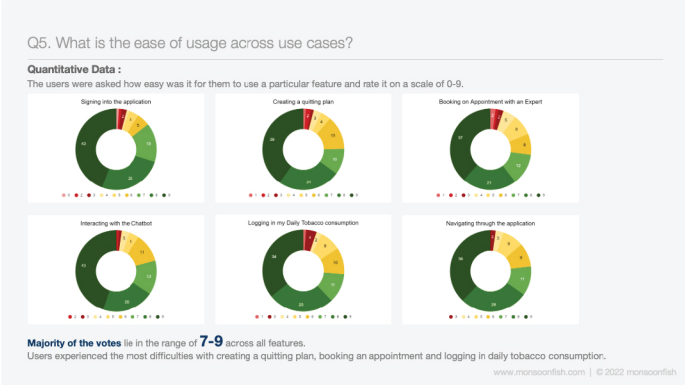


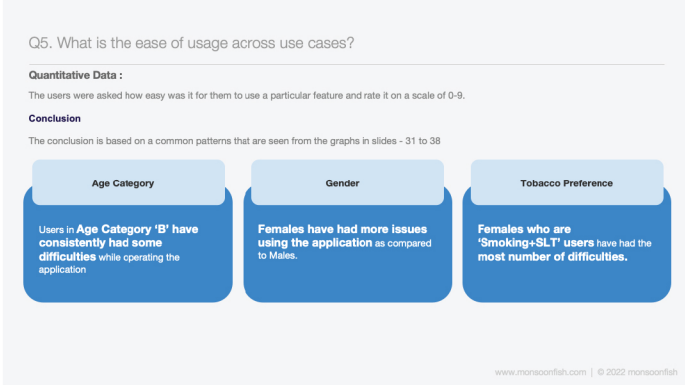


Qantitative Data : Task Analysis - Presentation

Similarly all the other tasks were also analysed:

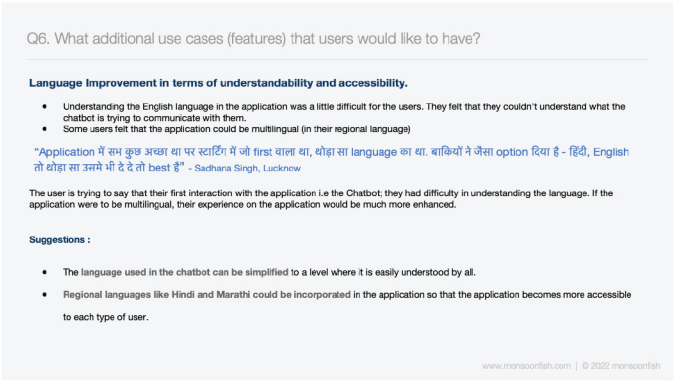
- Signing into the application
- Booking an appointment with an expert
- Interacting with the chatbot
- Logging daily tobacco consumption
- Navigating the application through infotainment content

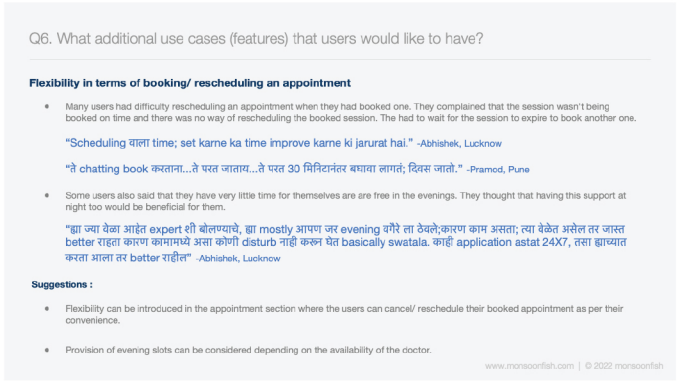


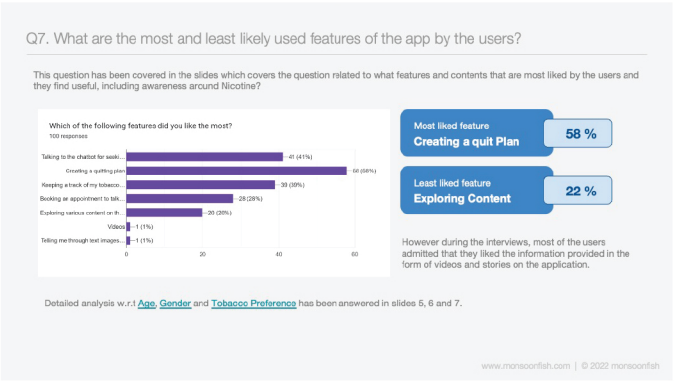


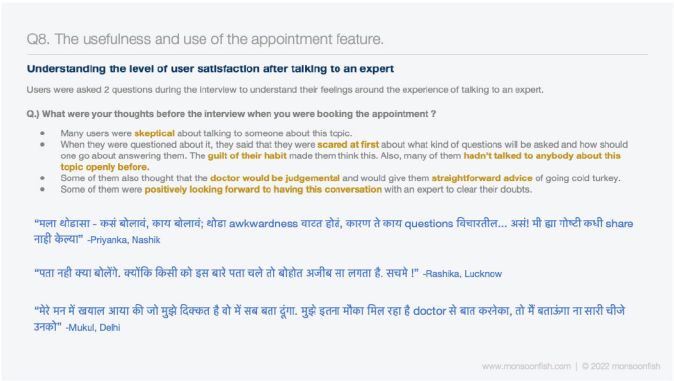
Qualitative Data : Presentation

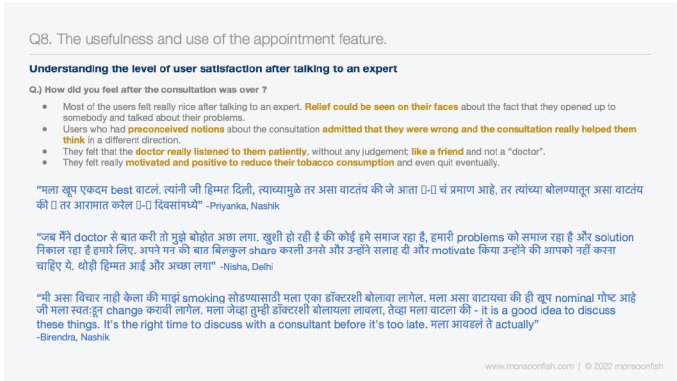
Data gathered from Google Forms and interview recordings was utilized to address the key research questions. To strengthen and substantiate the findings—especially those related to the usefulness and usage of all features—interview excerpts and the researcher's observations during the sessions were incorporated.

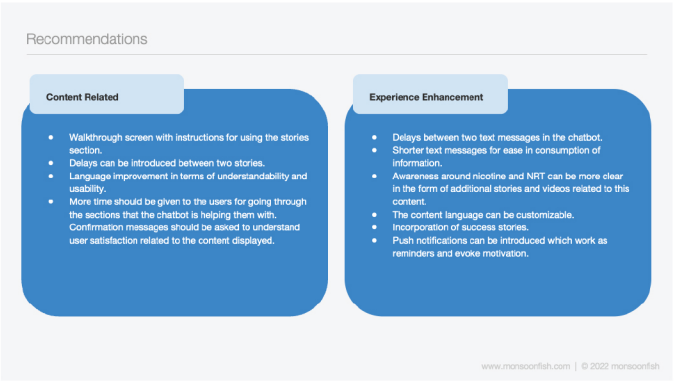












03

Domain Research for an Endoscopic Device

2022-23 | Formative Research | Ethnographic Research | Market Research

Project Overview

This project was conducted in collaboration with a multinational company specializing in the development of medical devices. The focus was on the design and development of a next-generation endoscopic device.The scope of the project involved extensive formative research aimed at understanding the context of use, user needs, and opportunities for design improvement. This included observational studies of current endoscopic devices in use across hospitals, where researchers closely examined workflows and identified pain points experienced by end-users.

As part of the ethnographic research, doctors, nurses, and technicians were interviewed to gain firsthand insights into their experiences. The research team also immersed themselves in real-time procedures within operating theaters, capturing observational data and understanding the practical challenges encountered during use. Additionally, a comprehensive market research study was carried out. This involved analyzing existing endoscopic devices in-depth—including their user manuals, form factors, and interaction patterns—to conduct a comparative study of similar products from different manufacturers. The findings from all research streams were synthesized into actionable insights, which were then presented to the client. These insights directly informed the design of the device and its user interface, ensuring that the final product was both user-centered and aligned with real-world clinical needs.

My Role

I worked as a co-researcher, collaborating closely with the design team at the company's main office. My responsibilities included conducting on-site visits, crafting and leading user interviews, and participating in immersive data collection within hospital settings. I independently conducted the market research and competitive analysis, and was also responsible for visualizing research findings for weekly presentations to clients. Additionally, I coordinated directly with cross-functional team members at the main office and took part in weekly review meetings with international clients.

Research Objectives

1. To understand the real-world clinical context in which endoscopic devices are used by observing live procedures in hospital operating theaters.
2. To identify key user needs, challenges, and pain points through in-depth interviews with doctors, nurses, and technicians involved in endoscopic procedures.
3. To analyze current workflows and device interactions in order to uncover usability gaps and areas for design improvement.
4. To conduct ethnographic research by immersing in clinical environments and capturing observational data that reflects actual usage scenarios.
5. To perform a comprehensive market and competitor analysis by studying existing endoscopic devices, their features, user manuals, and design elements.
6. To synthesize research findings into actionable insights that directly inform the design and user interface of a next-generation, user-centered endoscopic device.

User Research Goal

4. To conduct ethnographic research by immersing in clinical environments and capturing observational data that reflects actual usage scenarios.
5. Analyze the competitive landscape to identify gaps and opportunities, enabling the new device to deliver unique value beyond existing market offerings.
6. Inform the design of both physical and digital interfaces to ensure intuitive, efficient, and user-centered interactions.

Research Methodology + Objective

Immersive Observation: Observed real-time procedures in operating theaters to understand device use, workflows, and pain points.

Contextual Inquiry: Visited clinical settings to observe device usage, capturing behaviors and workarounds.

Task and Workflow Analysis: Mapped user interactions with devices to pinpoint inefficiencies and improvement areas.

Data Synthesis and Visualization: Transformed research findings into visual formats, like journey maps, for weekly client presentations.

User Interviews: Conducted semi-structured interviews with doctors, nurses, and technicians to gather insights on existing devices.

Market and Competitive Analysis: Analyzed user manuals, specifications, and features of competing devices to identify market gaps.

Research Study

Immersive Observation :

With prior permissions secured, we visited hospitals offering endoscopic procedures to conduct ethnographic research. Our study focused on the operating theatre environment, including the detailed placement and use of the endoscope and its controller. Data was meticulously captured through photographs and sketches, with careful attention to the sensitive nature of the setting. This immersive approach allowed us to document real-world interactions and contextual nuances, ensuring respect for ethical considerations and patient privacy throughout the process.

Layout of OT:

C-Arm positioned above the patient to gain X-ray images before procedure. The C-Arm is also used during Fluroscopy.

Olympus controllers connected to the Endoscope

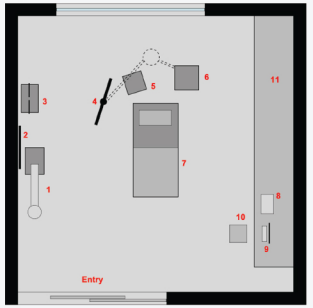
Position of Physician. The Physician was observed standing right next to the controllers

Monitors to view the endoscopy video while surgeon performs the procedure. This gives them a direct view from their standing position.

People present in the image: Person wearing gray scrubs is the technician, while person wearing green scrubs is the circulatory nurse. Patient is laid on the bed.

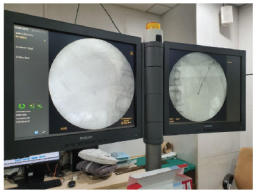
Ethnographic data Collection

Layout of OT:



1. C-Arm Apparatus
2. TV Screen
3. Cart with screens
4. Dual screens
5. Vitals apparatus
6. Cart with controller & processor
7. Patient's bed
8. Printer
9. Desktop for report generation
10. Chair
11. Counter with storage underneath

Fig 1:Plan of Procedure room

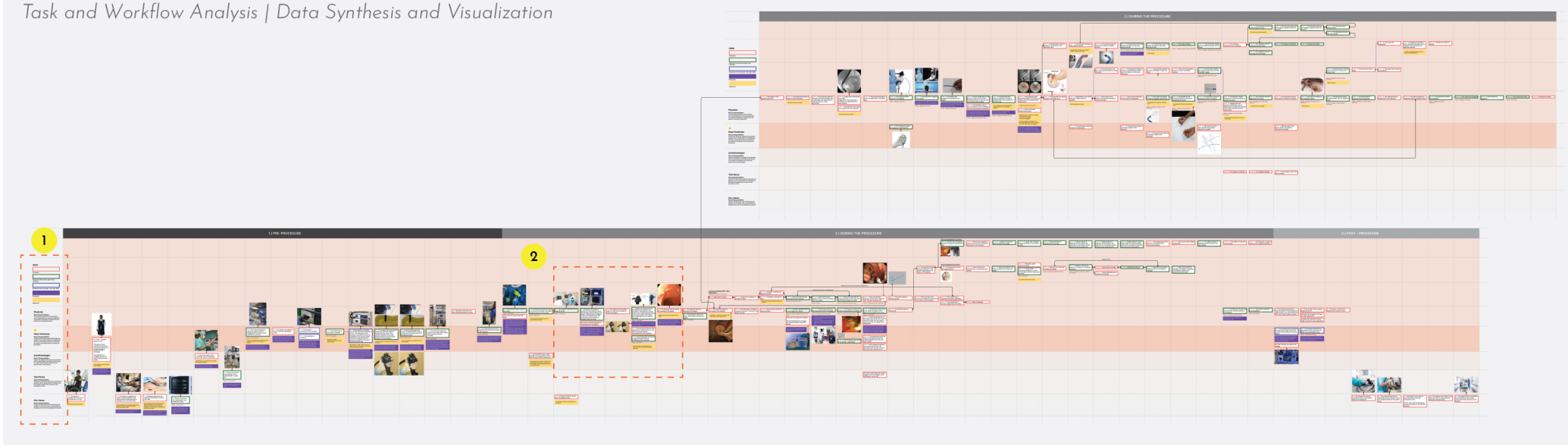


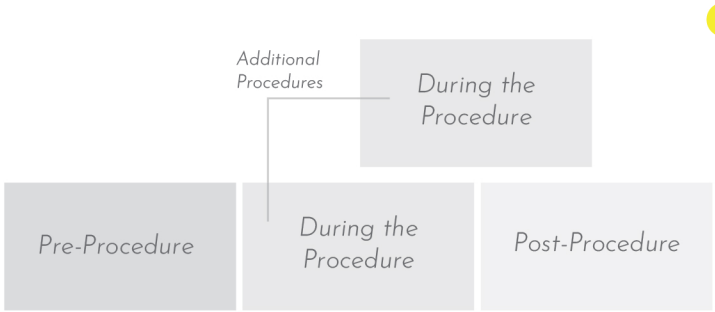
Images and Sketches Captured to Analyze the Operating Theatre Setting

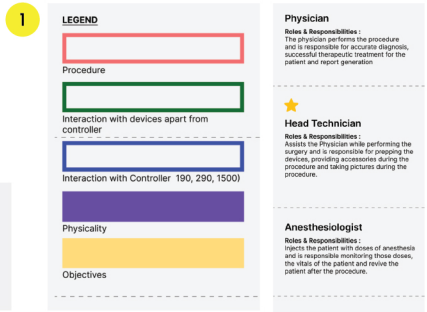
Ethnography : Immersive Observation | Contextual Inquiry |

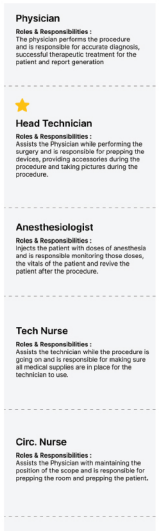
Following immersive observations of the endoscopic procedure, I created a comprehensive diagram illustrating the entire workflow, from pre-procedure preparations to post-procedure steps. This detailed diagram captures every minute aspect of the process, including additional procedural variations based on specific requirements, which were also monitored and documented. These insights directly informed the interface design, enabling feature prioritization and ensuring key functionalities were easily accessible based on the connected procedure.

Task and Workflow Analysis | Data Synthesis and Visualization



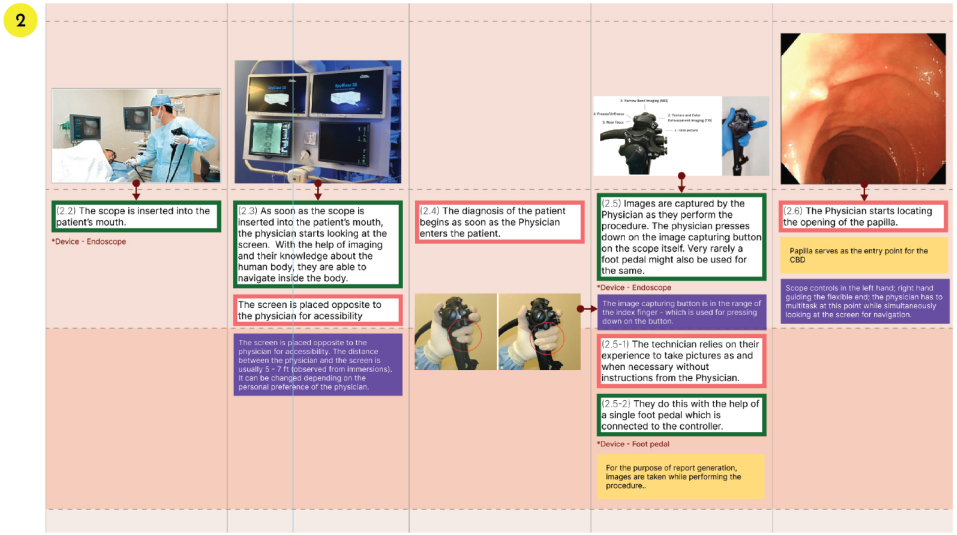






Key Diagram : Task and Workflow Analysis

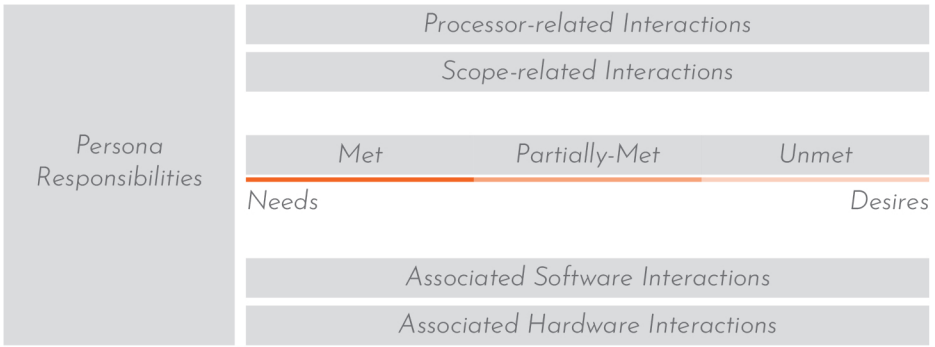
The legend accompanying the diagram outlines the key factors captured, including procedural steps and variables. It also identifies the individuals involved in the process, clarifying their roles and highlighting whose interactions with the controller and endoscope are most critical. This provides a comprehensive understanding of the workflow and informs design decisions for optimizing user interaction.



Details captured in the flowchart

User Interviews | Mapping needs and desires based on device interactions

Complex ethnographic data, gathered through immersions and interviews with doctors, nurses, and technicians, was synthesized into personas. These captured interactions with the endoscope and controller, integrated with qualitative interview data, and mapped onto a scale of needs and desires—distinguishing met, partially met, and unmet needs. Additionally, possible challenges within these needs and desires were identified, providing insights into opportunities for enhancing the user experience and pinpointing key areas of focus for designing an effective interface.



Key Diagram : Task and Workflow Analysis

Persona-Based Workflow Diagram

User Stories

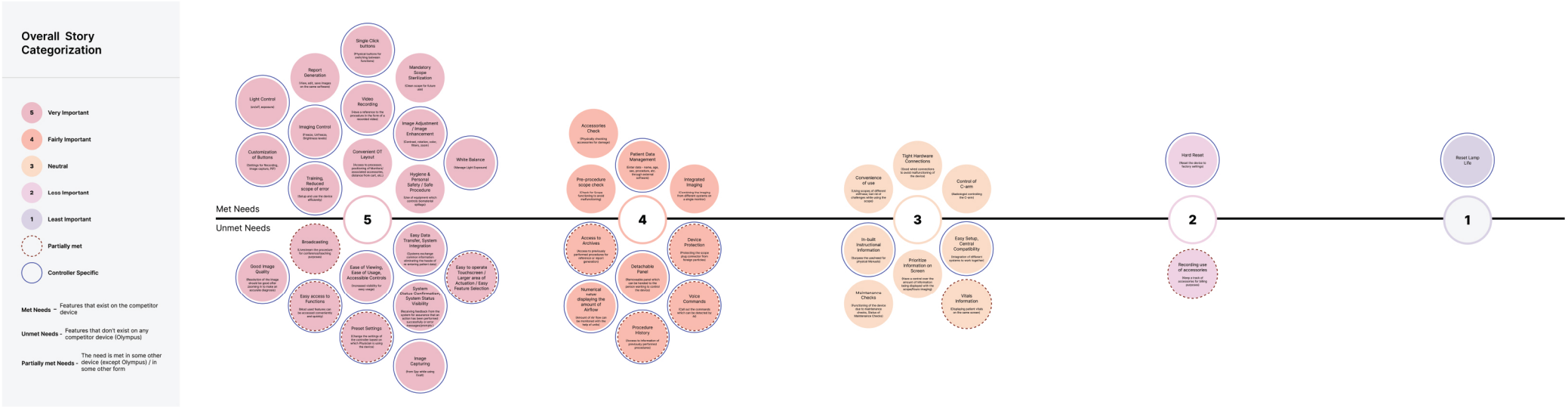
Mapping Met and Unmet needs on a priority scale

Data from user interviews and persona-based workflow diagrams was further refined by crafting user stories that highlighted the primary features used by device operators. Each feature was rated on a 1-to-5 scale, with 5 indicating the highest priority and most desired feature during the procedure.

This process pinpointed critical features for the user interface, incorporating user suggestions for enhancing functionality and improving the overall experience. By prioritizing these features, I developed a story categorization map that further distilled met, partially met, and unmet needs, guiding the design process effectively.

Sr.no	Phase	Role	Device	User Story	Ratings 1 - 5 (1 - Least Important, 5 - Most Important)	Needs	Type of needs	Feature	Devices on which the feature exists	Country
8	During Procedure	Physician	Monitors	As a Physician, I want to stand close to the Screen displaying the imaging from the endoscope so that I **can have a clear view of it without having to zoom into the image**	5	Convenient OT Layout	Observed need	High resolution picture quality after zooming in	-NA-	UK
9	During Procedure	Physician	Processor	As a Physician, I want to Recording, Image Capture and PIP settings to be part of the custom buttons so that i can **have easy access to these features during the procedure.**	5	Customization of buttons	Observed need	Recording, Image Capture and PIP as part of custom features	-NA-	UK, Germany
10	During Procedure	Physician	Processor, Scope	As a Physician, I want to be able to customize or use pre-programmed options for switching between PIP imaging, so that **I dont have to struggle while using these features during the procedure**	5	Customization of buttons	Specified need	Pre-Programmed options for different functions	-NA-	Germany
11	During Procedure	Physician	Processor	As a Physician, I want the touchscreen to be bigger in size so that I **can see the features properly without squinting my eyes.**	5	Ease of Viewing, Ease of Usage, Accessible Controls	Observed need	Big Touchscreen	-NA-	Germany
12	During Procedure	Physician	Processor	As a Physician, I want a simple solution to switch between Endo Screen and Spy Imaging while using PIP function so that **I dont have to go through multiple clicks on the processor to use the function**	5	Easy access to functions	Specified need	-NA-	-NA-	Germany
13	During Procedure	Physician	Monitors	As a Physician, I want to be able to switch between screens at the click of a button so that **i dont have to this manually reaching out to the screens.**	5	Easy Feature Selection	Observed need	One press buttons	-NA-	UK
14	During Procedure	Physician	Monitors	As a Physician, I want the image quality to be good so that i **can determine where a biopsy needs to be taken.**	5	Good Image Quality	Observed need	-NA-	-NA-	UK
15	During Procedure	Physician	Monitors	As a Physician, I want the monitors displaying the imaging from the Fluoroscope as well as the Endoscope close to me so that i can **view a clear picture and make an accurate diagnosis.**	5	Good Image Quality	Observed need	-NA-	-NA-	India
16	During Procedure	Physician	AutoCap RX	As a Physician, I want to use the AutoCap RX so that **there is minimal leakage of biomaterial**	5	Hygine & Personal Safety	Observed need	-NA-	-NA-	Germany

Spreadsheet : User Stories



Prioritization: User Stories

Research Study

Device Operation study

I conducted a thorough analysis of the installation and inspection workflows of key competitors' devices, diving deep into user manuals to understand their setup and operation within real-world settings. This study revealed discrepancies between the prescribed and actual use of the devices, shedding light on the most utilized features during procedures.

These insights were compiled and presented to clients, illustrating feature prioritization and usage patterns based on the operator's role.

Additionally, observational field notes and user interview data were documented in the presentations, providing critical insights that informed strategic design decisions.

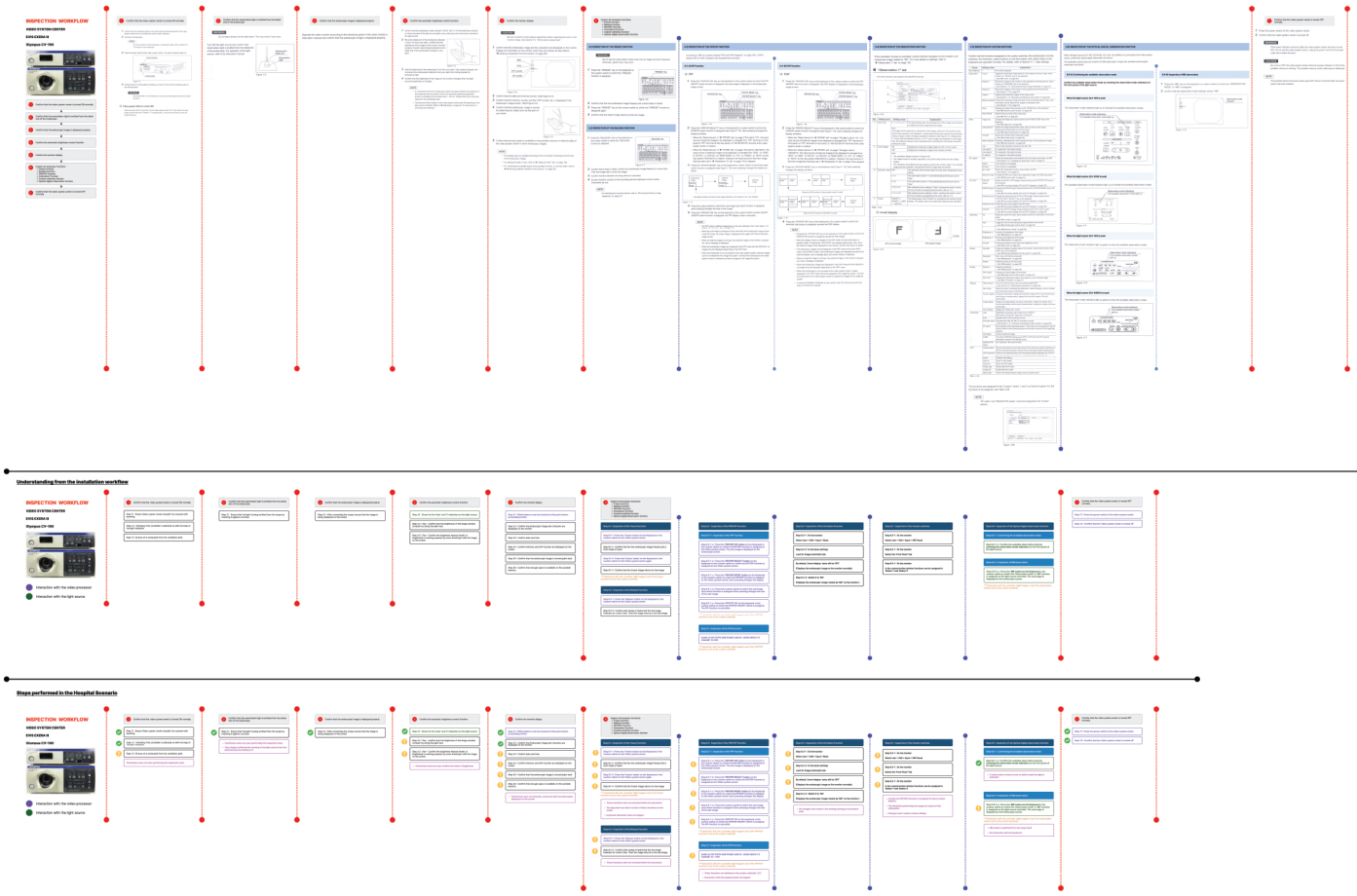
1.) Installation and inspection workflow

2.) Comparative Study: Recommended vs. Actual Workflow

3.) Researcher Insights: Most Used Features

4.) Researcher Insights: In Conjunction with User Interviews

1



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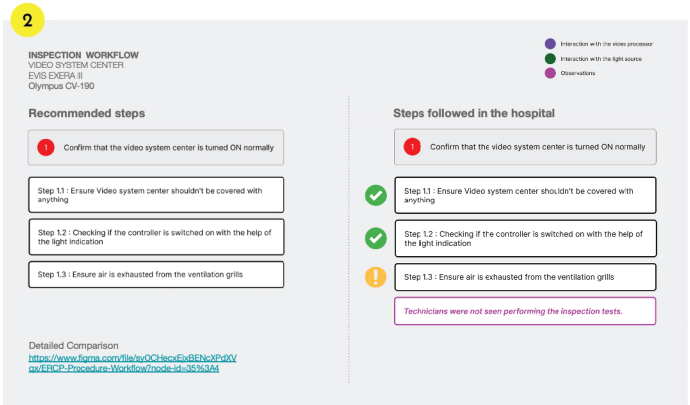

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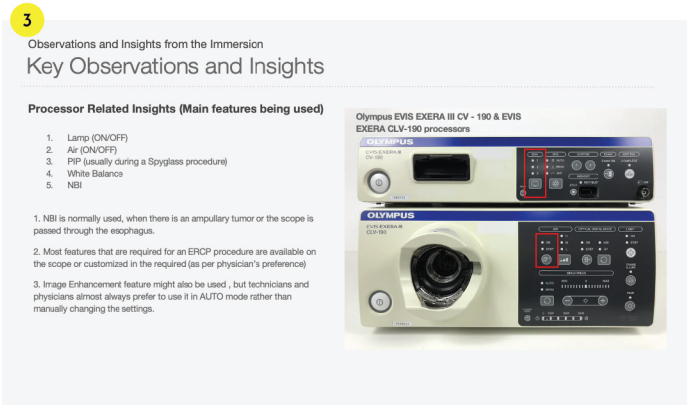

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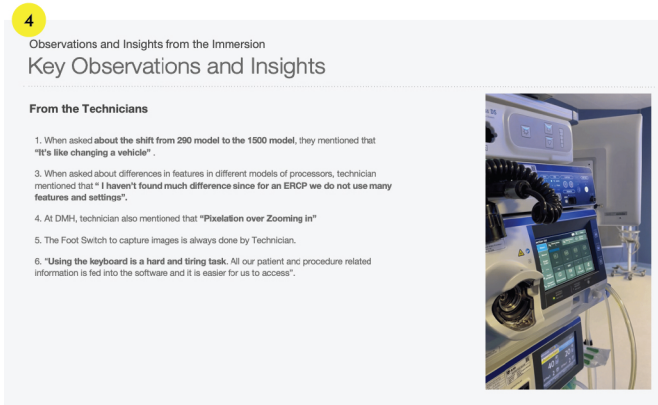

Competitor Study: In-Depth Feature-by-Feature Comparison

Major competitors of the device under development were identified and analyzed using user manuals and ethnographic research to evaluate the usability and utility of each feature. This approach pinpointed the most sought-after features in existing devices and uncovered opportunities for innovative enhancements to add value upon market launch. Features were also examined to determine if they could be operated directly from the endoscope, streamlining interaction by eliminating the need to access the main device. This narrowed down a set of high-priority features, optimizing efficiency and saving time during procedures. These insights were shared with the design team, enabling them to create a more efficient interface tailored to user needs.

Product	Sr.no	Feature	Images	Function	Objective	Operatiional from the scope switches
OlympusEVIS EXERA III	1	ENH - Enhancement Mode		- It electronically emphasizes the details and edges to increase sharpness - The 3 options for enhancement can be set as per the user's preference.	With this feature the Physcisian is able to get a better look at the area of focus (wrt structure,coloration) and make an accurate diagnosis. The enhancement mode allows for better picture quality. It is used for white light observations	TRUE
	2	IRIS Modes		3 Settings : Auto mode, Peak Mode, Average Mode -At any point of time either of the two settings can be assigned on the switch which can be a combination of Auto with avg, auto with peak and peak with avg - Auto mode is considered ideal for diagnosis as it is optimally suitable in adjusting the brightness of the endoscopic image. The auto iris modes can be selected using the "iris mode" switch on the front panel. Peak: The brightness is adjusted based on the brightest part of the endoscopic image. Average: the brightness is adjusted based on the average brightness of the endoscopic image.	Adjusting the brigtness of the endoscopic image	TRUE
	3	Custom Switches		- 2 custom switches on which any of the functions available could be assigned for direct activation at a press of a single switch. - Normally, PIP functions are assigned to the custom switches	The most used features can be set on the custom bottons. The settings can vary from user to user , this feature allows for ease in usage based on the user's preference.	TRUE
	4	Exam on		In CV-190, still images can be saved of procedures conducted in a patient-wise folder segmentation. During the procedure, with 'Exam on' activated, the technician presses release to save all the required images. After the completion of the procedure, the Exam on button is deactivated and the folder gets closed with all the images.	in the busy enviornment of the OT, the device is used constantly. for segregation of images taken during the procedure, this feature becomes useful as it saves images realted the the ongoing procedure. A seperate folder gets created for the individual patient allowign the hospital authorities to manage reports easily.	FALSE
	5	White Balance		The white balance mode is used to display the correct image colour on the monitor	The proximity of the light emited has an effect on the image seen on the monitor. White balance helps manage the colour to see the image with sufficient exposure.	FALSE
	6	Reset		Press and hold this button to return the settings changed during operation to the default settings	This feature allows the user to reset the device to factory settings for a fresh start.	FALSE
	7	Memory Port		For effective data management a slot for a portable memory device of 2gb in addition to the internal memory of 1gb is provided.	Portable external memory devices can be attched to the processor to store mode data.	FALSE
	8	PIP Comopsite Terminal		A composite video terminal can be connected to this terminal to input the images to be displayed together with the endoscopic images	Physicians may need visuals (from the scope and spyglass) for viewing two areas simultanously when they are conducting a procedure.	FALSE
	9	Video Connector port		Ensures backward compatibility with previous generation endoscopes such as CV-180 and the older ones. These are connected with a videoscope cable inserted into the port.	For making a conncetion between the scope and the processor	FALSE

CV-190, CLV-190 | CV-290, CLV-290 | Olympus 1500 | Pentax | Ambu® aView™ 2 Advance | Ambu® aBox™ Duodeno | All feature comparision